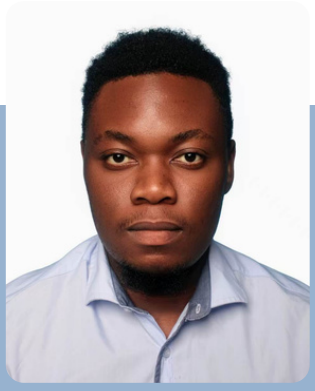
CHIPIKA ELTON



- Tianjin, China
- Phone: +86 130 3435 1546
- Email: 236140023@cauc.edu.cn
- GitHub: github.com/Maedza
- Portfolio: maedza.github.io/portfolio

PROFILE

Communication Engineering undergraduate with practical experience in protocol simulation and backend application development. Seeking a master's in Computer Science or Software Engineering, with a focus on distributed systems and reliable backend architecture.

Research interests: distributed systems; reliable, fault-tolerant backend architecture; communication protocols and networked applications

EDUCATION

B.Eng., Communication Engineering (通信工程) Sep 2023 – Jun 2027 (expected)

Civil Aviation University of China

- GPA: 3.15/4.0
- University Scholarship (Third Class), Civil Aviation University of China, 2024 – present
- Core coursework: Digital Communications (incl. computer networks), Digital Electronics, Signal Analysis (team project: audio-filtering application), Life-Critical Systems (elective, 90)

Whitfield Academy, South Africa, Maths and Sciences Track, 2018 – 2022

PROJECTS

YoBotz Jun 2025 – present

Conversational Automation Platform (Python, FastAPI, Redis, SQLite)

- Designed a layered, platform-agnostic architecture separating transport, orchestration, intent processing and business logic
- Built a FastAPI gateway with routing and delivery tracking; per-business secrets isolation and session storage on Redis
- Added catalog sync with cached offline fallback, thread-safe session management across concurrent users, and real-time owner notifications for new orders and bookings
- Deployed the private production implementation on Railway and functionally tested it on WhatsApp and Telegram
- Public showcase (YoBotz Lite, reduced Telegram-focused edition) : **github.com/Maedza/YoBotz_lite**

PROJECTS (continued)

GBN Lab May 2026 – present

Go-Back-N ARQ Simulator (Python, CustomTkinter)

- Implemented Go-Back-N ARQ: sliding window, cumulative ACKs, retransmission and RTT-based adaptive timeout
- Built a configurable noisy channel (bit error rate, packet loss) with live efficiency metrics
- Added step-by-step and frame-by-frame replay modes to inspect protocol behavior under failure
- **github.com/Maedza/GBN_Lab**

Micromouse: Autonomous Maze Robot (team of 3, in progress) Oct 2026 – present

- Main code contributor for ESP32 firmware (flood-fill maze solving, distance sensing); tested in an 8×8 maze, tuning stability and side-path detection for the 16×16 competition

SKILLS

- Systems: layered architecture, fault handling, concurrency, protocol design
- Languages: Python (primary), C, C++
- Tools: FastAPI, Redis, SQLite, Git/GitHub, Railway
- Spoken languages: English (fluent), Shona (native)